Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/31/2024
PI Name:	Tabassum, Shawana Ph.D.		
Project Title:	Leaf Sensor Network for In Situ and Multiparametric Analysis of Crop Stressors		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stabassum@uttyler.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	903-565-6461
Organization Name:	University of Texas, Tyler		
PI Address 1:	Electrical Engineering		
PI Address 2:	3900 University Blvd, Rbn 2005		
PI Web Page:			
City:	Tyler	**State:**	TX
Zip Code:	75799	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	03/01/2023	**End Date:**	02/28/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	O'Rourke, Aubrie	**Contact Phone:**	
Contact Email:	aubrie.e.orourke@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/28/2025 per NSSC information (Ed., 12/30/2024).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rajan, Nithya Ph.D. (Texas A&M University)		
Grant/Contract No.:	80NSSC23K0401		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project aims to advance the fundamental understanding of the hormonal responses in plants in a spaceflight-like environment through an in situ technology that collects and analyzes data on plant phytohormones in real time. Progressive variations in the phytohormone levels initiate a plant's defense mechanisms against environmental stressors. Salicylic acid, jasmonic acid, abscisic acid, and indole-3-acetic acid are among the most important regulators of induced defense mechanisms. Progressive variations in their levels have been reported under many drought and cold/heat-stressed conditions on Earth. However, the dynamic interaction mechanism of these hormones is not fully elucidated in a "space farming" setting due to the lack of technology, needed to facilitate in situ sensing. Real-time understanding of a plant's responses to stressors is essential to minimize stress-induced growth and yield declines in plants. Toward this end, this project develops a lightweight and integrated stem sensor with sensing elements to monitor plant hormonal variations in real-time. The system comprises a multiplexed hormone sensor for quantitatively measuring the primary defense hormones: Salicylic acid and abscisic acid. The impact of the following stressors on the hormone levels is analyzed: changes in temperature, lighting, and growth media. In contrast to the traditional discrete, disruptive, in vitro, time-intensive, and heavyweight instruments used for molecular analysis, our proposed sensor is plant-wearable and provides in situ monitoring capabilities. We have developed functional correlations of the measured hormonal variations with physiological indicators (photosynthesis, respiration, and transpiration) to differentiate the effect of growing conditions on individual plant responses. The knowledge gained from this project will advance future research on predicting and improving plant growth and productivity under spaceflight stressors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study aims to enhance our understanding of a plant's defense mechanisms against a combination of environmental stressors by providing real-time data on variations in the plant's defense hormones. Accurate and real-time measurements of plant parameters are expected to provide agricultural producers with immediate insights into crop stress levels, facilitating precise adjustments in irrigation and fertilizer application. The outcomes of this project are poised to have a lasting positive impact on terrestrial agriculture by: (1) providing a tool for early assessment of crop water and nutrient uptake, (2) contributing to the existing and new databases on irrigation and fertilizer management, modeling data, and linking local data with national-level databases, (3) developing science-based irrigation and fertilizer management policies, and (4) improving the environmental sustainability of agricultural practices.
Task Progress:	Our central hypothesis is that real-time data collected on the phytohormone levels in plants can be used to determine the fundamental ways plants interact with microclimatic conditions associated with living in space. The expected deliverables of this research are the development of technology and a dataset that could then be used to monitor plant responses and identify the most productive growing conditions in space. In this regard, we have made the following progress and accomplishments during the reporting period. We have developed an electrochemical sensor for monitoring the levels of salicylic acid (SA) and abscisic acid (ABA) in the plant stem. The sensor design is devised considering that cowpea possesses a dicot stem, leading to the arrangement of vascular bundles along the periphery. Also, the diameter of cowpea plants ranges from 5mm to 10mm. Consequently, the active region of the sensor penetrates through the vascular bundle, leaving the remaining portion accessible for signal acquisition. The sensor was calibrated for the entire concentration range of SA and ABA typically found in plants. Like animals, plants produce hormones as signaling molecules to control various growth processes. SA and ABA are two primary stress hormones in plants. Progressive variations in their levels have been reported in many drought, salt, and cold/heat-stressed plants. We have tested the sensors in a growth chamber facility in June 2024 and a greenhouse in November 2024. Cowpea plants were grown under controlled conditions in coir pith, hydroponic solutions, and a mixture of peat and perlite media. We used a completely randomized design, with 3 replications used for each treatment. We observed that plants grown in hydroponics had the highest assimilation, transpiration, stomatal conductance, and CO2 concentration, which indicates high water use. In contrast, plants grown in the peat perlite media demonstrated the lowest levels for these parameters, thus indicating water stress. Meanwhile, plants grown in coir pith media displayed intermediate values for these metrics. These results strongly support our sensor measurements. It was noted that the plants grown in the peat perlite media exhibited the highest concentrations of both SA and ABA, with SA concentration reaching upto 8.1 µM and ABA concentration reaching upto 28 µM, indicating a water stress scenario. In contrast, the plants grown in the hydroponic solution exhibited the lowest concentrations of both SA and ABA, with SA and ABA concentrations peaking 0.195 µM and 4.65 µM, respectively, indicating sufficient water use by the plants. Meanwhile, plants grown in coir pith media displayed intermediate concentrations for both the hormones, which indicates moderate stress. The above and below-ground plant biomass data further corroborates the plant physiological traits and hormonal responses. Our immediate research plan includes the following: • Complete statistical analysis correlating plant physiological traits with the hormonal responses. • Complete validating the SA and ABA measurements with the sensors against the values obtained from high-performance liquid chromatography (HPLC).
Bibliography Type:	Description: (Last Updated: 08/06/2025)
Abstracts for Journals and Proceedings	Pali M, Islam MN, Poudyal C, Wheeler W, Rajan N, Tabassum S. "SMART-PH: Sensing Modality to Assist Real-time Tracking of Plant Health in spaceflight-like conditions." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Poudyal C, Wheeler W, Pali M, Islam MN, Tabassum S, Rajan N. "Exploring the phytohormonal responses of plants for space agriculture ising in situ sensor technology." CANVAS 2025 (ASA, CSSA, SSSA International Annual Meeting), San Antonio, TX, November 10-13, 2024. Abstracts. CANVAS 2025 (ASA, CSSA, SSSA International Annual Meeting), San Antonio, TX, November 10-13, 2024. , Nov-2024

Abstracts for Journals and Proceedings	Poudyal C, Wheeler W, Pali M, Islam MN, Tabassum S, Rajan N. "Real-time monitoring of plant stress hormones for extra-terrestrial agriculture." Data Science Space Week: From Earth to Mars, Observations in Biological and Physical Sciences, College Station, TX, April 1-5, 2024. Abstracts. Data Science Space Week: From Earth to Mars, Observations in Biological and Physical Sciences, College Station, TX, April 1-5, 2024. , Apr-2024